

AJMAL ABBAS

Machine Learning Engineer | Generative AI | LLM | Agentic AI

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[Portfolio Website: https://skillful-lilac-ec7.notion.site/AJMAL-A-3a4ba46fe55a4b93908e876a9920e435](https://skillful-lilac-ec7.notion.site/AJMAL-A-3a4ba46fe55a4b93908e876a9920e435)

Professional Summary:

Machine Learning Engineer and AI Researcher with expertise in designing, training, evaluating, and deploying AI/ML solutions across marine ecosystem modeling, cybersecurity, computer vision, and trustworthy multimodal AI. Proven experience developing end-to-end machine learning pipelines, surrogate modeling systems, deep learning architectures, and predictive analytics solutions from research to deployment. Published researcher with IEEE and DSN contributions, including work on confidence-aware AI systems and multimodal learning. Skilled in Python, PyTorch, TensorFlow, LLM-powered applications, model optimization, evaluation, and scalable data-driven solutions. Passionate about applying AI to solve complex real-world challenges across diverse domains.

Skills and Certificates

- **Data Science & AI:** Pytorch, TensorFlow, SciKit-Learn, Machine Learning, Deep Learning, Stats Techniques, NLP, Big-Data, Hugging Face
 - **Programming and Tools:** Python, R, SQL, PySpark, Tableau, Django, Minitab, ARC GIS, GeoPandas, Lang flow
 - **Domains:** Cybersecurity, Agriculture, Marine Ecosystem, Material Modeling & Business
 - **Certifications:** Snowflakes, AI Automation and Agentic AI, **Azure Certificates:** Synapse Analytics, Data Bricks, OpenAI, Data Engineering
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Education:

Master of Science in Data Science with 4.0/4.0 CGPA

Sep 2024-May 2026

University of Massachusetts Dartmouth, MA, USA

- **Master's Thesis:** [Emulating Marine Ecosystem Dynamics: A Machine Learning Approach for Biomass Prediction and Forecasting](#)

Post Graduate Program in Data Science with 3.7/4.0 CGPA

Sep 2023-May 2024

Vellore Institute of Technology (VIT), Bangalore, India

- **Project Worked and Published Title:** A Cutting-Edge Deep Learning Models for Paddy Leaf Disease Detection and Classification

Published in IEEE Publication [3rd IEEE International Conference on Artificial Intelligence for Internet of Things \(ALLOT\) \(Acceptance Rate 8%\)](#)

Bachelor's in Data Science with 3.8/4.0 CGPA

Sep 2021-May 2023

The American College, Madurai, India

- **Industry Internship & project:** Completed a Full Stack development and data Analytics internship at Desyndev , Applying software development and data science techniques to real-world applications.
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Work Experience:

[Graduate Research Assistant - School of Marine Science and Technology \(SMAST\)/UMASS Dartmouth, MA, USA](#) Sep 2025 – May 2026

- Formulated and implemented a hybrid ML-DL ecosystem emulator (Random Forest + LSTM) to predict and forecast biomass for 51 marine species using 58 years of NEUS Atlantis simulations, achieving 94% overall accuracy on highly zero-inflated, heterogenous ecological data.
- Engineered a high-performance surrogate modeling pipeline with optuna- optimized Random Forest (0.89s inference) and LSTM forecasting (0.26s inference), enabling near-instant scenario evaluation and long-horizon forecasts to 2035, dramatically reducing computational cost versus full ecosystem simulations.

[Graduate Research Assistant – College of Engineering /UMASS Dartmouth, MA, USA](#)

Jun 2025- Aug 2025

- Architected and created a hybrid software vulnerability detection framework combining AST stylometry features with CodeBERTa-small-v1 embeddings, enabling scalable evaluation across Devign, MVD and ReVeal datasets and achieving 80% classification accuracy and packaging the system as a reusable python library(hybrid-vul) for practical vulnerability analysis.
- Streamlined end to end preprocessing and training pipelines by benchmarking CPU vs GPU feature extraction and demonstrating faster MLP training using CPU-based features (2.61 vs 7.76s), enabling efficient deployment on low-power, resource -constrained systems.

[Graduate Research Assistant- College of Engineering /UMASS Dartmouth, MA, USA](#)

Dec 2024 - Jan 2025

- Implemented and prepared a deep learning-based Network Intrusion Detection (NIDS) leveraging Graph Neural Network (GNNs) to model complex network traffic, detect anomalies and identify cyber threats in real time, achieving 97% detection accuracy.
 - Integrated AI-driven security strategies using state-of the art GNN and deep learning models, improving threats detection accuracy and system scalability in dynamic network environments through close collaboration with peers and faculty.
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Project Experience:

[From Weak Embeddings to Trustworthy Multimodality: Modality-Agnostic Learning with Confidence-Aware Fallback \(Published in SIEDS 2026 \)](#)

- Deployed unified multimodal learning pipeline (text-image-video) using contrastive embedding, integrating CIFAR-100 ResNet, CLIP, BLIP captioning and R3D-18 video models to produce calibrated confidence scores across modalities.
- Improved model trustworthiness through probability calibration (ECE, temperature scaling), lowering calibration error and supporting confidence- driven multimodal decision fallback.

AI Policy Intelligence Platform

- Designed an agentic AI research pipeline using Langflow and LLM-Powered workflows to automate state level Ai policy discovery, evidence extraction, structured synthesis and claim verification across government and public information sources.
- Implemented evidence-grounded validation, source citation, confidence-based verification, structured Database storage, change detection and an interactive dashboard for comparing legislation programs, funding and AI initiatives across states.

GeneGuard AI- Agent AI for Genetic Disk Risk Intelligence

- Developed Gene Guard AI, a genetic disease risk intelligence workflow integrating family history with Clinvar and GWAS genomic evidence, enabling structured risk analysis across Lynch syndrome, familial hypercholesterolemia, hypertrophic cardiomyopathy and Brugada syndrome through family -history normalization and disease-specific genetic evidence analysis.
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Publications:

- [VulStyle: A Multi-Modal Pre-Training for Code Stylometry-Augmented Vulnerability Detection \(DSN 2026 Acceptance rate 19%\)](#)